

Shreerang Vaidya

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 <https://s-mv.github.io/>

Computer Science undergraduate interested in compiler design, machine learning, embedded systems and distributed systems.

Education

Vivekanand Education Society's Institute of Technology

2022 – 2026

B. E. in Computer Engineering — Aggregate GPA: 9.13

Relevant courses: Systems Programming and Compiler Construction, Theoretical Computer Science, DSA, Microprocessors, Digital Logic and Computer Architecture, Operating Systems, OOPM, DBMS, Probabilistic Graphical Models, Quantitative Analysis, Discrete Structures and Graph Theory, etc.

- 2nd runner up in BuzzPro, 2024.
- Special mention in Hack-AI-thon 2.0, 2024.
- 3rd runner up in Awakening The Scientist, 2022.

Technical Skills

- **Languages:** C, C++, Rust, Python, Go, Java, JavaScript
- **Technologies:** LLVM, PyTorch, TensorFlow, MERN Stack, FastAPI, WebAssembly
- **Tools:** Unix/Linux, Git, GDB, Valgrind, Make/CMake, Docker, Selenium
- **Concepts:** Systems Programming, Compiler Design, OS, Embedded Systems, DSA, ML/DL, Web Development, CI/CD

Experience

Accelus Robotics

May 2025– July 2025

Intern

- Developed a bytecode VM in C++ extending the rs274ngc NIST specification for efficient CNC machine control.
- Forked the LinuxCNC repository in order to introduce the extension in its ecosystem.
- Extensively applied compiler design, OS, embedded and systems programming concepts.

Banao Technologies

Jan 2024– Jun 2024

Web Scraping Intern

- Learned the applications of web scraping, DBMS, backend development, etc. through various projects.
- Gained valuable insights about the internal workings of a startup and team collaboration.

4thventure IoTspace

Jun 2020– May 2021

Intern

- Worked on a project to develop a user-facing React application.
- Collaborated on various frontend and backend systems.

Notable Projects

gcode++ | — <https://github.com/s-mv/gcode-plus-plus>

- Bytecode VM implementation extending the NIST rs274ngc g-code interpreter, adding conditions, looping, variables for versatility.
- Extensively applying compiler design, systems programming, as well as data structures and algorithms.

smvm | — <https://github.com/s-mv/smvm>

- Register-based Virtual Machine in C with assembler. Release v0.1.1 available, CI/CD with GitHub Actions.
- Utilizes DSA, microprocessors, compiler design principles. Helped me learn Github Actions CI/CD.

Black-Scholes Visualizer | — <https://github.com/s-mv/black-scholes-visualizer> (run it here)

- Educational tool to demonstrate the Black-Scholes pricing model for European options using Rust + WebAssembly.
- Implements the full mathematical model including $\Phi(x)$, and d_1 , d_2 terms with interactive visualization.

smvrt | — <https://github.com/s-mv/smvrt> (work in progress)

- Minimal preemptive real-time operating system written in C and assembly, targeting ARMv7-based hardware like Raspberry Pi and QEMU.
- Includes cooperative multitasking, custom scheduler, basic UART CLI, and memory-mapped I/O with no external dependencies.
- Bootable image runs bare-metal without Linux, showcasing low-level systems design.

Achievements

- **AIR 1627** in GATE CSE 2025 certifying strong fundamentals.
- Co-hosted the 1st Godot India Community Game Jam (GICGJ), 2025.
- Co-authored the paper "**KisanSevak: Enhancing Crop Management and Market Forecasting through Machine Learning**": <https://jet-m.com/wp-content/uploads/129-JETM8261.pdf>
- Submitted an entry for IICT Segfault Hackathon – DiviniC: A modern Holy C Compiler with JIT Runtime and Lazy Error Handling.
- 2 time Google Code-In Participant (2017-18).